# 1. Inception Score (IS)

The Inception Score attempts to quantify how realistic and diverse the generated images are. A higher score is better.

## What it Measures:

1. **Image Quality (Clarity/Realism):** Is the image clearly recognizable as a specific object (e.g., a bird, a car)?
2. **Sample Diversity (Variety):** Does the generator produce many different types of images, or does it suffer from **mode collapse** (generating only a few types)?

$$\text{IS}(G) = \exp\left(E_{\mathbf{x}\sim p_g}\left[D_{KL}\left(p(y \mid \mathbf{x}) \parallel p(y)\right)\right]\right)$$

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|---|---|---|
| $\mathbf{x} \sim p_g$ | A generated image sample ($\mathbf{x}$) drawn from the Generator's distribution ($p_g$). | The expected value is calculated over many generated samples. |
| $p(y \mid \mathbf{x})$ | The conditional probability distribution of labels ($y$) given a generated image ($\mathbf{x}$), obtained from the pre-trained Inception V3 classifier. | **Clarity/Realism:** We want this distribution to be **peaked** (low entropy), meaning the classifier is very confident it knows what the image is. |
| $p(y)$ | The marginal probability distribution of labels ($y$) across **all** generated samples. | **Diversity:** We want this distribution to be **uniform** (high entropy), meaning the generator is covering all classes equally. |
| $D_{KL}(P \parallel Q)$ | The Kullback-Leibler Divergence, measuring the difference between two distributions. | If the difference between $p(y \mid \mathbf{x})$ and $p(y)$ is **large** (high KL divergence), the model is generating both clear and diverse images, resulting in a **high IS score**. |

# 2. FID Score (Fréchet Inception Distance)

The FID Score is the current state-of-the-art metric for evaluating generative models. It measures how similar the *distribution* of generated images is to the *distribution* of real training images. A **lower score is better.**

## What it Measures:

The FID measures the **distance** between two data distributions:

1. The feature distribution of the **real training data**.
2. The feature distribution of the **generated samples**.

$$\mathrm{FID}(p_{\mathbf{r}}, p_g) = \|\boldsymbol{\mu}_{\mathbf{r}} - \boldsymbol{\mu}_g\|_2^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_{\mathbf{r}} + \boldsymbol{\Sigma}_g - 2\left(\boldsymbol{\Sigma}_{\mathbf{r}}\boldsymbol{\Sigma}_g\right)^{1/2}\right)$$

| | | |
|---|---|---|
| $p_{\mathbf{r}}$ | The distribution of features for **real** images. | |
| $p_g$ | The distribution of features for **generated** images. | |
| $\boldsymbol{\mu}_{\mathbf{r}}$ and $\boldsymbol{\mu}_g$ | The **mean** feature vectors (calculated from the Inception V3 embedding layer) for the real and generated data, respectively. | **Fidelity:** The $\|\boldsymbol{\mu}_{\mathbf{r}} - \boldsymbol{\mu}_g\|_2^2$ term penalizes the model if the *average* generated image is far from the *average* real image. |
| $\boldsymbol{\Sigma}_{\mathbf{r}}$ and $\boldsymbol{\Sigma}_g$ | The **covariance matrices** of the feature vectors for the real and generated data. | **Diversity/Mode Collapse:** The second term (involving the trace, $\mathrm{Tr}$) penalizes the model if the generated data lacks the same variation or spread (covariance) as the real data. |